Multi – Agent Insurance System Report

Course Name: AAI

Institution Name: Medicaps University – Datagami Skill Based Course

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Project Number : AAI - 20

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Academic Year: 2025-2026

Problem Statement & Objectives -

1. Problem Statement:

The insurance sector is inherently complex, involving a wide variety of policies, technical jargon, and conditional benefits that are often difficult for the average user to interpret. While digital transformation has enabled users to access insurance information online, most existing platforms are limited to static data presentation and lack intelligent, personalized guidance.

Users commonly face challenges such as:

- Difficulty in understanding insurance terminology like premiums, deductibles, coverage limits, and exclusions
- Inability to compare multiple plans effectively based on their specific needs
- Lack of real-time assistance when making decisions related to insurance purchases
- Overwhelming amount of information without structured guidance

Moreover, conventional chatbot systems used in this domain are often rule-based or rely solely on large language models without proper grounding. This leads to:

- Generic or inaccurate responses (hallucinations)
- Lack of domain-specific reliability
- Absence of actionable outcomes such as retrieving actual insurance plans

Another critical limitation is the tight coupling between AI systems and backend databases, which can lead to security risks and poor system modularity.

Hence, there is a strong need for a robust, intelligent, and scalable insurance advisory system that:

- Understands user intent dynamically
- Provides context-aware and domain-grounded responses
- Assists users in selecting appropriate insurance plans
- Ensures safe, validated, and reliable AI interactions
- Integrates seamlessly with real-world backend systems

This project addresses these challenges by leveraging Agentic AI, Generative AI, and Retrieval-Augmented Generation (RAG) within a modular and secure architecture.

2. Project Objectives:

The main objective of this project is to design and implement a Multi-Agent Insuranc Advisory System that delivers intelligent, reliable, and user-centric insurance guidance through advanced AI techniques and full-stack integration.

Primary Objective:

To develop a production-ready AI system capable of understanding user queries, retrieving relevant knowledge, and assisting in insurance decision-making through a multi-agent framework.

Specific Objectives:

- To design an intent-aware system that can classify user queries into different categories such as explanation, plan retrieval, and purchase assistance
- To implement a multi-agent architecture where specialized agents handle distinct responsibilities, such as fact extraction and response generation, ensuring modularity and efficiency
- To integrate Retrieval-Augmented Generation (RAG) for improving response accuracy by grounding outputs in domain-specific knowledge
- To enable dynamic interaction workflows (Buy Flow) that guide users step-by-step in selecting suitable insurance plans
- To establish a secure separation between AI and backend systems, ensuring that AI components do not directly access databases
- To incorporate validation and guardrails, including input validation, domain restriction, and prompt injection protection, for safe AI operation
- To support multi-turn conversational capabilities using session-based memory for maintaining context
- To design a unified response structure that supports multiple output formats such as text, UI cards, and optional visual elements
- To ensure scalability and performance optimization using asynchronous processing and lightweight models

3.Scope of the Project:

The scope of this project encompasses the design and development of a comprehensive AI-powered insurance advisory platform that integrates modern AI techniques with a full-stack application architecture.

In Scope:

- Development of a multi-agent system using frameworks like CrewAI to enable task specialization and collaborative decision-making

- Implementation of intent-based query routing, allowing the system to dynamically select appropriate workflows (Explain, Buy, Get Plans)

- Construction of a RAG pipeline using vector embeddings and similarity search (FAISS) to retrieve relevant insurance knowledge
- Integration with a backend system (Node.js, Express, MongoDB) for fetching real insurance plans and related data
- Deployment of an AI service layer using FastAPI to handle asynchronous communication and agent orchestration
- Implementation of session and memory management to support contextual, multi-turn user interactions
- Design of a frontend interface (React with Zustand) capable of rendering structured responses including text and plan cards
- Incorporation of robust validation and guardrails, such as:
    - Input sanitization
    - Domain-specific query filtering
    - Prompt injection detection
    - Output structure validation
- Support for extensible features, such as optional image generation and multimodal responses

Out of Scope:

- Direct integration with live insurance companies, underwriting systems, or payment gateways
- Legal verification or compliance auditing of insurance policies
- Handling of sensitive personal data such as medical records or financial history for personalization

- Deployment in large-scale distributed cloud environments with auto-scaling infrastructure
- Advanced recommendation systems based on deep user profiling or behavioral analytics

Proposed Solution -

1.Key features:

 Intent-Based Routing: Identifies user intent (Explain, Buy, Get Plans) and directs queries accordingly.

 Multi-Agent Architecture: Uses Researcher and Advisor agents for structured fact extraction and clear explanations.

 RAG Integration: Retrieves relevant insurance knowledge to generate accurate, context-based responses.

 Interactive Buy Flow: Supports step-by-step user interaction for selecting suitable insurance plans.

 Backend Integration: Connects with MERN stack to fetch real insurance data.

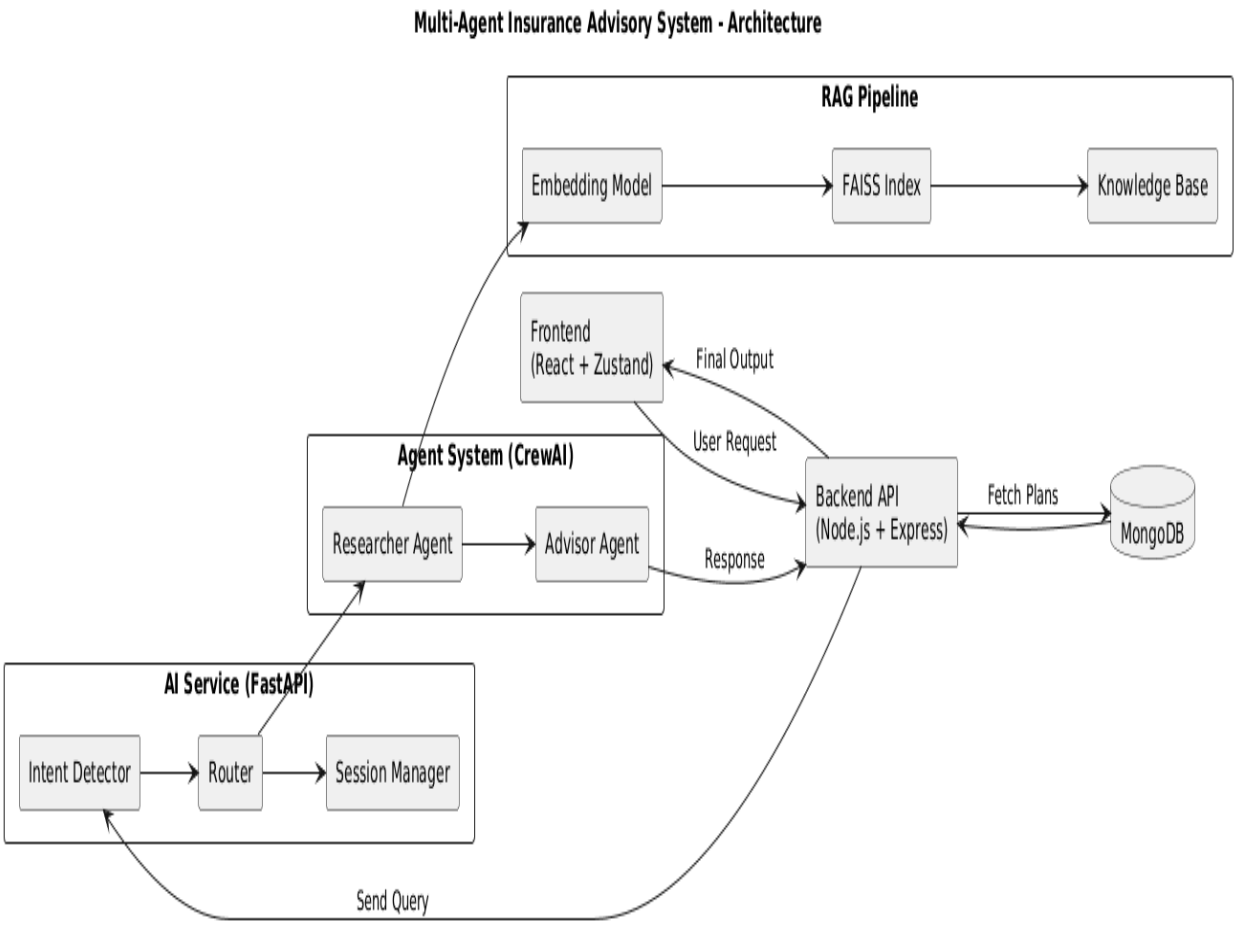
 Security & Guardrails: Includes input validation, domain restriction, and prompt injection protection.

 Session Memory: Maintains conversation context for multi-turn interactions.

 Unified Response Format: Provides structured outputs (text + UI cards) for frontend display.

2.Overall Architecture / Workflow:

- User sends request via frontend (React)
- Backend (Node.js) receives and forwards query to AI service
- AI Service (FastAPI) performs intent detection and routing
- Query is processed by Researcher & Advisor agents
- RAG pipeline retrieves relevant knowledge using FAISS
- Backend fetches insurance data from MongoDB (if needed)
- Advisor agent generates final response
- Backend sends response to frontend for display



3.Tools & Technologies Used :

- **Frontend:**
  - React (Vite)
  - Zustand (State Management)
- **Backend:**
  - Node.js
  - Express.js
- **Database:**
  - MongoDB (NoSQL database for storing plans and user data)
- **AI Service Layer:**

- FastAPI (Python for asynchronous AI processing)

- **Agent Framework:**
  - CrewAI (for multi-agent system implementation)
- **Large Language Model (LLM):**
  - Gemini API (Gemini 2.5 Flash)
- **RAG Components:**
  - Sentence Transformers (all-MiniLM-L6-v2) – for text embeddings
  - FAISS – for vector similarity search
- **Optional Tools:**
  - Hugging Face – for image generation / future extensions

Results & Output -

 The system successfully understands user queries and classifies them based on intent (Explain, Buy, Get Plans).

 It generates accurate and context-aware responses using the RAG pipeline, reducing incorrect or generic answers.

 The multi-agent system effectively separates:

- Fact extraction (Researcher Agent)
- Response generation (Advisor Agent)

 The system supports interactive buy flow, guiding users step-by-step in selecting insurance plans.

 Real-time data is fetched from the backend (MongoDB), ensuring practical and usable outputs.

 The application handles multi-turn conversations using session-based memory.

 Responses are delivered in a structured format including:

- Text explanations
- Insurance plan cards (UI output)

 Implemented guardrails ensure safe and reliable responses by preventing invalid inputs and prompt injection attacks.

 The system demonstrates scalable and modular architecture, suitable for real-world deployment.

1.Screenshots/Outputs

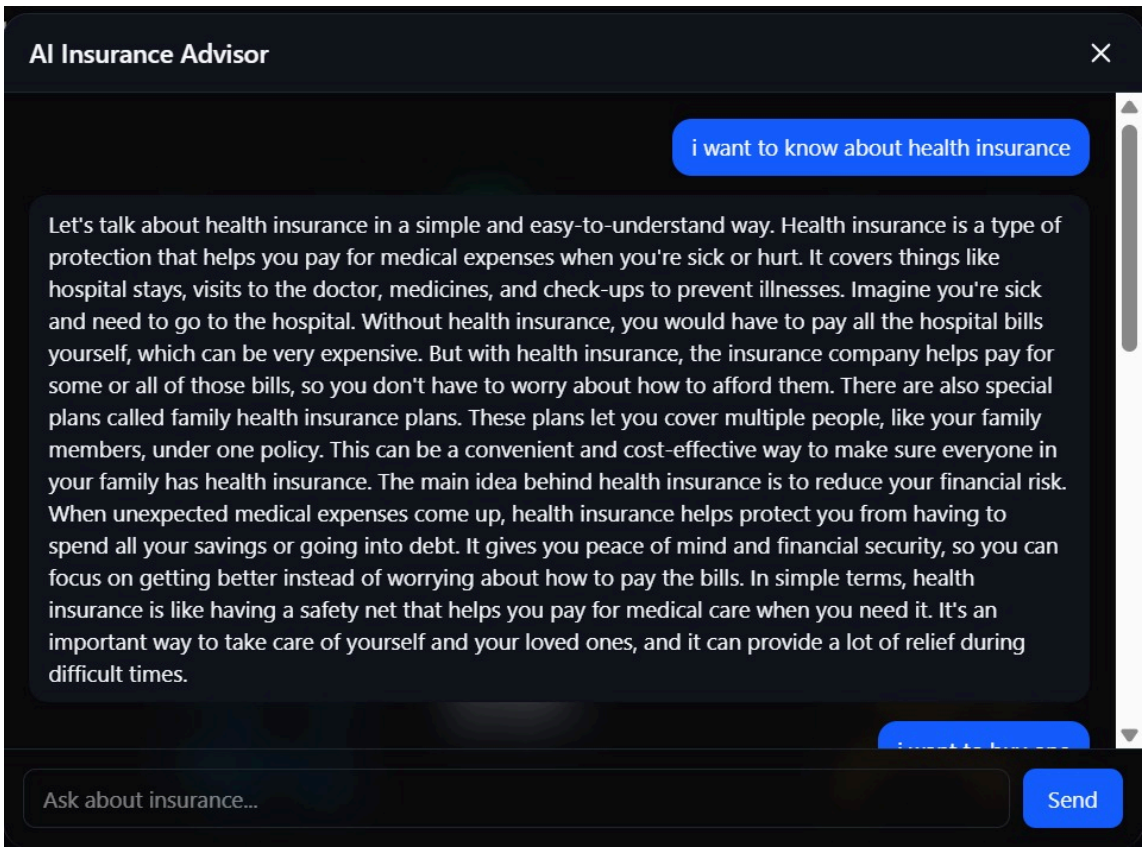


Fig1.1

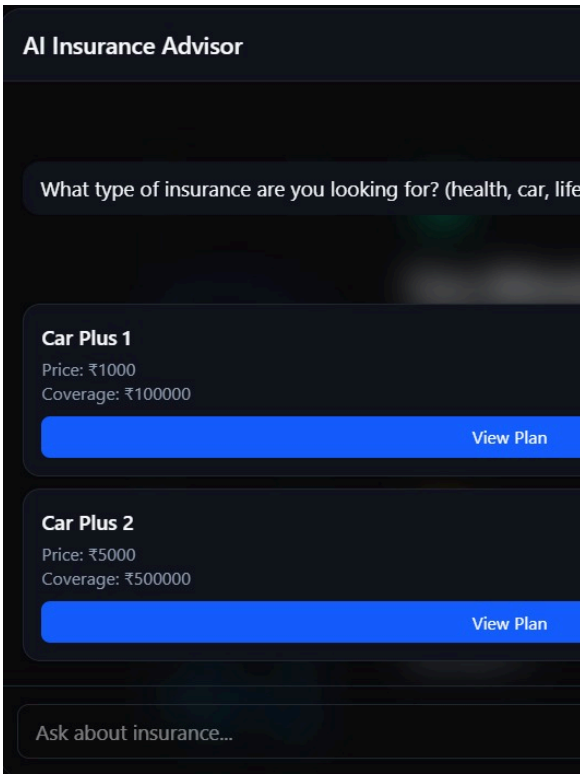


Fig 1.2

2.Reports/Dashboards/Model

Reports

- Generates structured outputs for user queries in a unified format
- Includes:
  - Text-based explanations
  - Insurance plan recommendations
- Provides clear and concise summaries of insurance policies
- Ensures consistency and readability for end-users
- Can be extended for analytics and performance reports

Dashboard

- User-friendly interface built using React
- Displays:
  - User queries and responses
  - Recommended insurance plans (cards view)
- Supports real-time interaction with the AI system
- Maintains session-based conversation history
- Designed for easy navigation and better user experience

Models

- LLM Model:
  - Gemini 2.5 Flash – for natural language understanding and response generation
- Embedding Model:
  - Sentence Transformers (all-MiniLM-L6-v2) – for converting text into vector form
- Vector Search Model:
  - FAISS – for similarity search and knowledge retrieval
- Agent Model:
  - CrewAI-based multi-agent system (Researcher & Advisor agents)

3.Key Outcomes

 Successfully developed a multi-agent AI system for insurance advisory.

 Achieved accurate intent detection and query routing for different user needs.

 Implemented RAG-based knowledge retrieval, improving response accuracy and reducing hallucinations.

 Enabled interactive insurance plan selection through dynamic buy flow.

 Integrated AI with real backend (MERN stack) for practical data retrieval.

 Ensured secure and reliable system behavior using validation and guardrails.

 Supported multi-turn conversations with session-based memory.

 Delivered structured and user-friendly outputs (text + UI cards).

 Built a scalable and modular architecture suitable for real-world applications.

Conclusion –

The Multi-Agent Insurance Advisory System presents a comprehensive and intelligent solution to the challenges associated with understanding and selecting insurance policies. By integrating Agentic AI, Generative AI, and modern full-stack technologies, the system moves beyond traditional static platforms and delivers a dynamic, interactive, and user-centric experience.

A key strength of the system lies in its intent-driven architecture, which enables accurate classification of user queries and ensures that each request is processed through the most appropriate workflow. The adoption of a multi-agent framework further enhances the system's effectiveness by separating responsibilities between the Researcher Agent and Advisor Agent, thereby improving both the quality and

reliability of the responses.

The incorporation of Retrieval-Augmented Generation (RAG) plays a crucial role in grounding the system's outputs in domain-specific knowledge. By leveraging vector search through FAISS and semantic embeddings, the system minimizes hallucinations and ensures that responses are both relevant and factually accurate. This significantly improves user trust and system credibility.

Another important aspect of the system is its integration with a real backend (MERN stack), which allows it to retrieve and present actual insurance plans. This bridges the gap between theoretical AI responses and practical application, making the system more useful in real-world scenarios. The clear separation between the AI service and the backend database also ensures better security, scalability, and maintainability.

The implementation of guardrails and validation mechanisms strengthens the robustness of the system by preventing invalid inputs, restricting domain-specific queries, and protecting against prompt injection attacks. Additionally, the use of session-based memory enables smooth multi-turn interactions, allowing the system to maintain context and provide more personalized assistance.

From a system design perspective, the use of FastAPI for asynchronous processing, along with lightweight embedding models, ensures efficient performance and responsiveness. The

unified response schema further enhances frontend compatibility by supporting structured outputs such as text and UI elements.

In conclusion, the project successfully demonstrates how advanced AI techniques can be combined with scalable system architecture to build a reliable and production -ready insurance advisory platform. The system not only improves user understanding and decision-making but also establishes a strong foundation for future enhancements such as personalization, real-time integrations, and multimodal capabilities.

Future Scope & Enhancements –

- Integration with Real Insurance Providers:
Extend the system to connect with live insurance company APIs for real-time policy data and transactions.

- Advanced Personalization:
Incorporate user profiling (age, income, preferences) to provide more tailored insurance recommendations.

- Voice-Based Interaction:

Add speech-to-text and text-to-speech capabilities for a more interactive and accessible user experience.

- Multimodal Capabilities:
Enhance the system to support images, charts, and visual explanations for better understanding of policies.

- Mobile Application Development:
Develop a dedicated mobile app (Android/iOS) for wider accessibility and user engagement.

- Improved Recommendation Engine:
Integrate machine learning models to suggest optimal plans based on user behavior and historical data.

- Scalability Enhancements:
Deploy the system on cloud platforms with microservices architecture for handling large-scale users.

- Enhanced Security Mechanisms:
Implement advanced authentication, encryption, and compliance with data protection standards.

- Multi-language Support:
Enable support for multiple regional languages to reach a broader audience.

- Continuous Learning System:
Improve the AI model using user feedback and interaction data for better performance over time.